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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/093,967	01/06/2023	Yu Fang	SONA-0012-U01-C06	1957

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GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

EXAMINER	
NGUYEN, NGA X	

ART UNIT	PAPER NUMBER
3662	

NOTIFICATION DATE	DELIVERY MODE
03/31/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 18/093,967	Applicant(s) Fang et al.	
	Examiner NGA X NGUYEN	Art Unit 3662	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03/07/2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,3-4 and 6-18 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,3-4 and 6-18 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/21/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

The current application relates:

- CON in part of application No. 17/833614, and 17/195589 (Pat. No. 11,538287).
- Provisional applications No. 63/197021 (filed Jun. 4, 2021), 62/986444 (filed Mar. 6, 2020) & 63/024383 (filed May 13, 2020) & 63/123531 (filed Dec. 10, 2020)

Response to Arguments

2. Applicant's arguments filed 03/07/2025 have been fully considered and are moot in view new grounds of rejection. Applicant's arguments with respect to the claims have been considered. The arguments do not apply to any of the references being used in the current rejection.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

3. Claim(s) 1-18 is/are rejected under 35 U.S.C. 103 as being unpatentable over Lagnemma (20190179304) in view of Kamdar (20050187680).

With regard to claim 1, Lagnemma discloses an apparatus, comprising:

a first controller external of a vehicle **(a computer process 171 is external of vehicle 10, see [0037]+), comprising:**

an automation definition circuit structured to:

implement an automation user interface for a device external of the vehicle **(a teleoperator 13 located outside the vehicle, at a location remote from the vehicle, implementated a supervisory control, see [0037]+);**

provide at least one automation recipe to the automation user interface, wherein each automation recipe comprises a trigger detection description defining trigger conditions for initiating an automated action and an automation action description defining the automated action to be performed when the trigger conditions are met **(the supervisory control asserted by supervisory commands 191 provided by the human operator, see [0037], the supervisory command could express a short-term movement objective such as a changing the vehicle's lane of travel on a road network, and etc., see [0038]+, and a human operator makes a section using a mouse, speaking the objective or selection, touching a touch sensitive display screen, see [0040]+);** and

interpret an automation request in response to user communications with the automation user interface and based at least in part on the at least one automation recipe **(the supervisory commands are selected by a human operator who can be a passenger in a vehicle or located remote from the vehicle, see]0037]. The**

human operator makes the selection using a mouse, speaking the objective or selection, touching a touch sensitive display screen to achieve an objective, see [0040]+);

wherein the first controller further comprises:

an automation management circuit structured to determine an automation description in response to the automation request **(the supervisory commands provided by the computer process 171 (see Fig.1) expresses objectives to be attained in operation of the vehicle, a sequence of roads to be followed to reach a destination or a turn to be made from denominated road to another according to a planned route nor expressed of control inputs to the accelerator, brake, and etc., see [0037]-[0038]+);** and

an automation command circuit structured to provide the automation description to a second controller disposed onboard the vehicle **((the supervisory commands provided by the computer process 171 (see Fig.1) to vehicle's control system of the vehicle 10 with input selection from operation in vehicle or input command, see [0099]+)**

wherein the second controller is configured to implement an automated vehicle response, responsive to the automation description **(the supervisory control process takes as an input the desired lane or path of travel and generates a sequence of control actions used by actuator to guide the vehicle through the desired lane of travel, see [0103]+).**

Lagnemma is silent about selecting a group of automation recipes in response to a user characteristic of a user, and to provide the at least one automation recipe in response to the selected group of automation recipes

Kamdar discloses a system for providing automated vehicle (see the abstract). The system includes a mobile vehicle communication unit 110, a telematics unit 120 which communicates with the vehicle via communication network 112, client computers 150, a service center 170, see [0012]+. The service center 170 receives user reference data from user computer 150 to set and stores personalization setting supplied by the client for automation of door lock, unlock behavior, radio station preset selections, climate controls, and other services, see [0024]+ which meets the scope of **"selecting a group of automation recipes in response to a user characteristic of a user, and to provide the at least one automation recipe in response to the selected group of automation recipes".**

It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to modify Lagnemma by including receiving user's references data which is set and stored personalization setting for vehicle's automation for preset selections and controlling as taught by Kamdar. The combination of Lagnemma and Kamdar for an adapted system for improving the performance in the vehicle's automation.

With regard to claim 3, Lagnamma teaches that the apparatus of claim 1, wherein the automation request comprises a selected one of the at least one automation recipes

(the computer system automatically generates a candidate trajectory set and execute a selected trajectory of the set, see [0056]+).

With regard to claim 4, Lagnamma teaches that the apparatus of claim 1, wherein each of the at least one automation recipe comprises at least one of: an automated operation of the vehicle; a portion of an automated operation of the vehicle; or a template for an automated operation of the vehicle **(the supervisory commands involves selecting a travel lane, turning onto a different roadway, changing velocity, stopping travel, and etc., see [065]+).**

With regard to claim 6, Lagnamma teaches that the apparatus of claim 1, wherein the automation description comprises at least one parameter description that is a vehicle specific parameter description (**automated candidate trajectory generation ensure that the driving rules e obeyed by the ego vehicle in complex scenarios involving relevant driving rules or the presence of numerous obstacles, see [0064]+).**

With regard to claim 7, Lagnamma teaches that the apparatus of claim 1, wherein the automation command circuit is further structured to provide the automation description to a third controller disposed onboard a third vehicle (**a communication device 60 communicates the measured or inferred or both properties of other vehicles' states and conditions, The devices include V2V, see [0070]+).**

With regard to claim 8, Lagnamma teaches that the apparatus of claim 1, wherein the automation description is for at least a cabin lighting of the vehicle (**activate turn signals, hazard lights, and etc., see [0079]+).**

With regard to claim 9, Kamdar teaches that the apparatus of claim 1, wherein the selected trajectory automation description is for at least a window of the vehicle **(sends signals to various units of equipment and system within MVCU 110 to perform various function such as opening the trunk, door, and etc., see [0014]+).**

With regard to claim 10, Kamdar teaches that the apparatus of claim 1, wherein the automation description is for at least a smart phone input for the vehicle **(in-vehicle mobile phone 134, see [0016]+).**

With regard to claim 11, Kamdar teaches that the apparatus of claim 1, wherein the automation description is for at least a seat of the vehicle **(personal setting for automation such as door lock/iunlock, climate controls, custom button configurations which interprets of seat button configuration, and etc., see [0024]+) .**

With regard to claim 12, Kamdar teaches that the apparatus of claim 1, wherein a trigger is at least one of a completion of a task, a detection of an event, a trigger evaluation, or an authorization (**upon reception of a command from a user interface,**

executes the primary diagnostic script at a predetermined time interval, or when a predetermine event occur, see [0037]+).

With regard to claim 13, Kamdar teaches that the apparatus of claim 1, wherein the trigger detection description defines an aggregated data condition calculated from multiple vehicle parameters or a statistical analysis of vehicle data collected over time **(trigger event data is collected from multiple vehicle at any service center, see [0041]+).**

With regard to claim 14, Lagnamma teaches that the apparatus of claim 1, wherein the automation action description defines at least one of: branching action sequences with conditional logic, or cyclical actions to be repeated until a condition is met (**the selected supervisory command is provided in the control system 21 of the self-driving vehicle uses algorithms to determine a sequence of steering wheel, brake, and etc., see [0041]+).**

With regard to claim 15, Lagnemma teaches that the apparatus of claim 1, wherein multiple automation recipes can be active simultaneously, and wherein the second controller is structured to manage conflicts between concurrent recipes based on priority values assigned to each recipe (**the input device allows an operator located inside the vehicle or at a remote location to directly command the position of the vehicle steering wheel, brake, and throttle, select the gear, and activate turn signals, see [0079]+).**

With regard to claim 16, Lagnamma teaches that the apparatus of claim 1, wherein the trigger conditions include at least one time delay or hold period after the trigger conditions are initially met before initiating the automated action **(database includes programs for operating telematics 220. The programs stores triggers at triggers 234 at predetermined event occurs, see [0037]+).**

With regard to claim 17, Kamdar teaches that the apparatus of claim 1, wherein the automation user interface presents available automation recipes according to the user characteristic, wherein the user characteristic comprises at least one of a user roles, an entity associated with the user, a user authorization level, or a user's historical interaction patterns with the automation user interface **(the client computer sends user preferences through a web page interface, see [0021]-[0022]+).**

With regard to claim 18, Kamdar teaches that the apparatus of claim 1, wherein the trigger condition is at least one of user control of cabin temperature, lighting, infotainment, seats, windows, sunroof, cabriolet top, driving mode, adjustment of any other actuator or vehicle interface in response to voice commands, smart phone inputs, buttons in the vehicle, or detected vehicle operating conditions or events **(personal reference settings from a client at computer 150 such as door lock/unlock, climate controls, and etc., see [0024]+).**

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to NGA X NGUYEN whose telephone number is (571)272-5217. The examiner can normally be reached M-F 5:30AM - 2:30PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, JELANI SMITH can be reached on 571-270-3969. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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NGA X. NGUYEN
Examiner
Art Unit 3662

/NGA X NGUYEN/
Primary Examiner, Art Unit 3662